Asli Awaaz

असली आवाज़

A call-security layer for bank helplines that scores every call for voice-clone risk and, when the risk is high, asks for one more proof instead of blocking the customer.



Detectors look for generator patterns.

Illustration, not data.

Team

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RAKSHAM, AI Cybersecurity Hackathon (IIT Delhi with Amazon)

Round 1 submission. Challenge statement 01: Voice Cloning Fraud Detection

A cloned voice on the helpline

Before the call
The fraudster takes a short voice sample from a public video or voice note, and personal details from a leaked database.

The call
They ring the helpline as the account holder, answer the identity questions and speak in the cloned voice.

The approval
Voice ID, or the agent's ear, accepts the voice. The answers are correct, so a limit change, new payee or SIM-linked change goes through.

After the call
The money moves. The customer finds out from an SMS or a failed payment.

₹22,845.7 crore

lost to cyber fraud in India in 2024, up from ₹7,465.2 crore in 2023; 22.68 lakh cybercrime incidents on NCRP.

All cyber fraud, not only voice

One precedent

In 2023 a reporter used an AI clone of his own voice to pass his UK bank's voice ID and reach his account (Vice).

Primary user
Helpline agent and the fraud-ops analyst behind them

Protected
The account holder whose voice is cloned

In scope
Helpline calls at banks that use or add voice ID, as ICICI did in 2015

Not in scope
Calls to relatives, video deepfakes, SIM-swap itself

Typical pattern, not one case. Sources: Lok Sabha USQ 344, 22 Jul 2025; Vice, Feb 2023; ICICI, May 2015

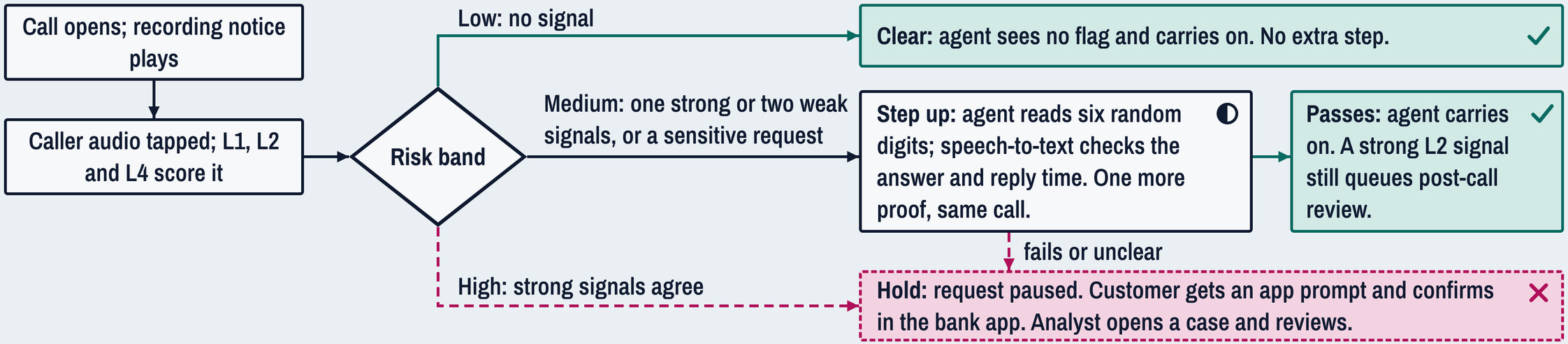
Where Asli Awaaz changes a helpline call

Today Trust rests on what the caller knows and how the voice sounds.

Customer calls the helpline → Menu routes the call to an agent → Agent asks identity questions → Agent approves the request ✕

With Asli Awaaz The call keeps its first three steps. The score runs beside the call; the band reaches the agent within 10 s of the caller's first reply.

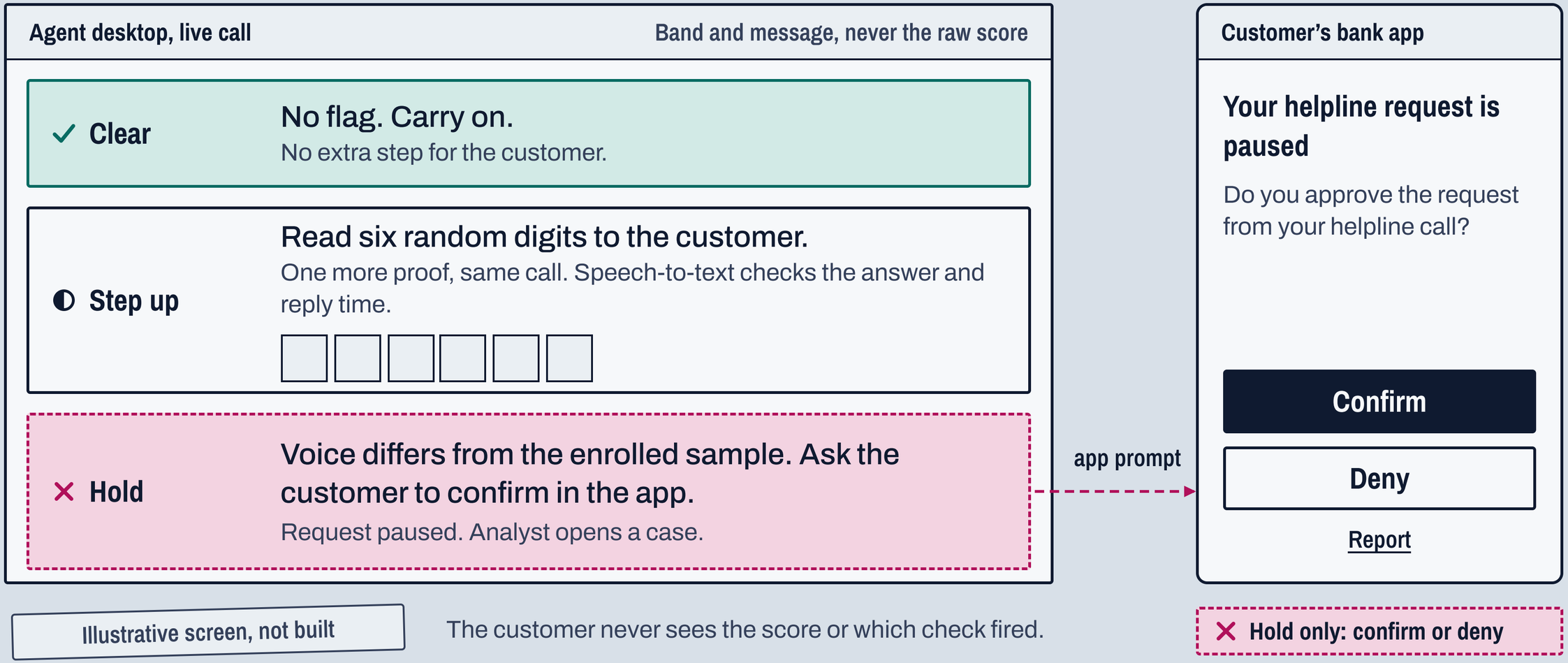
Target, not a result



Every path writes an evidence record: band, layer scores, model versions, actions taken and who decided.

Illustrative: the Today lane is typical. No bank data was used.

What the agent and the customer see



Messages as on pages 3 and 8. No customer data shown.

Three scores set the band; two checks follow

Check	Method	Does not catch
L1 Speaker match Does the voice fit the enrolled customer?	SpeechBrain ECAPA-TDNN, cosine score; enrolled customers only	A clone good enough to pass the match
L2 Synthetic-speech check Does the audio carry generation artefacts?	wav2vec2 XLS-R 300M with an AASIST head; weaker AASIST-L fallback on CPU	Generators it has not seen; heavy codec damage
L4 Call context Does the request look like fraud?	Rules: new device or payee, amount jump, urgency words	A fraudster who stays inside normal behaviour
Run only after the band is set		
L3 Digit challenge Is a live person answering a random prompt?	faster-whisper speech-to-text plus reply time	A live voice changer, or fast type-to-speech
L5 App confirmation Does the account holder approve?	Push prompt in the bank app; analyst queue	A customer whose phone is also compromised

Clear ✓
No strong or weak signal. No extra step.

Step up ◐
One strong or two weak signals, or any sensitive request: limit, payee, mobile or device change.

Hold ✕
Strong signals agree, or the digit challenge fails or is unclear. Policy pauses the request; customer confirms in the app.

Not enrolled? L2 and L4 still run; risky requests go to app confirmation.

Strong or weak: past a threshold set on held-out data; Platt-calibrated scores.

What a helpline call needs from a detector

Detectors trained on ASVspoof 2019 lost accuracy on real-world audio. In one study the error rate grew by up to about

1,000%

Relative increase in equal error rate (Müller et al., Interspeech 2022, on the In-the-Wild dataset).

Phone calls add codec loss and packet loss on top. We will train and test on that audio.

	Detector tested on benchmarks	Asli Awaaz
Audio	Clean studio clips	8 kHz phone audio with G.711, AMR, GSM, Opus and packet loss
Language	English benchmarks	A planned Hinglish test set, recorded with consent
Output	A score	A band, a plain message and the next step
When unsure	Pass or block	Step up: ask for one more proof
Limits	Often unstated	Listed for each layer (page 5)
Where it runs	After the call, on an uploaded file	Beside the live call, inside the bank network

Why it matters next to OTP and app confirmation. A helpline call that changes the registered mobile or device can switch those controls off. Asli Awaaz guards that call.

Closest product. Pindrop Pulse, a commercial live check for cloned voices in contact centres. Its page claims 99% on known engines and over 90% on unseen ones. We have not tested it.

Typical, not a survey of products. Sources: Müller et al., Interspeech 2022; Pindrop product page

Tests will use phone-line audio

Dataset	Used for	Terms
ASVspoof 2019 LA	Train and baseline for L2	ODC-By
ASVspoof 5	New attacks, codec conditions	ODC-By 1.0
In-the-Wild	Real-world check	Apache 2.0 or CC BY-SA 4.0; sources differ
Own Hinglish set	Accent and code-mix test; planned 4 speakers × 20 clips	Written consent per speaker, including clone tests

What we report

Target, not a result

Equal error rate

Per codec and loss condition. Pass line: beat the ASVspoof 2019 baseline in every codec; the number comes after the first run.

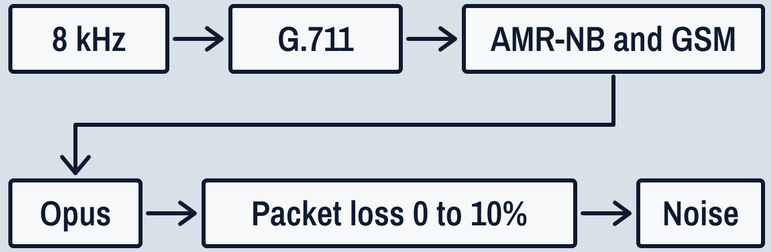
Genuine step-ups

Per accent group, at a fixed alarm rate. Target: under 5%.

Time to first band

Target: 10 s or less from the caller's first reply.

Phone-line degradation, applied with ffmpeg



Top three of six assumptions (Appendix B)

A1 The bank can fork caller audio from its platform.

A2 8 kHz audio keeps enough signal for L1 and L2.

A4 Public data stands in for fraud calls.

Datasets: ASVspoof 2019 LA, ASVspoof 5, In-the-Wild (Müller et al., 2022)

Trust: five questions from the brief

Designed, not yet tested

What data do we need?	Live caller audio and its transcript, a consented voice embedding, and bank flags: new device, new payee, amount jump. Not collected: account numbers or balances; no audio or transcript kept unless a case is held.
What if the system is wrong?	Nothing is blocked on one weak signal. A false alarm costs the customer one extra step. A miss may be caught by post-call review and the bank's payee and limit controls. Overrides are logged with a reason.
Who can act on the result?	The agent sees a band and a message, not raw scores. Policy can pause a request; only an analyst can release or escalate it. Account actions stay with the bank's fraud process.
How is the result explained?	A plain message with the next step, such as "Voice differs from the enrolled sample. Ask the customer to confirm in the app." A score alone is never shown as proof.
How does it resist misuse?	Digits are random; attempts are capped per account and per call. The caller learns only that an extra proof was asked, never which check fired or the score. Evidence is role-gated and logged. Voiceprints cover consenting customers only.

Another way in. A customer who declines voice enrolment verifies in the app only. Voice is never the only way to verify.

Fairness. Step-up and Hold rates for genuine callers by accent group, including digit errors on Hindi numerals. Allowed gap: 3 points.

Design questions follow the Round 1 brief, "Build for Trust".

How a bank adopts it

each step needs the bank's sign-off

Assumption: RBI and DPDP fit

Phase 1: Shadow

Scores calls, shows nobody. Scores are compared with confirmed fraud outcomes.

Ends when false Step-up rates are under target in every accent group.

Phase 2: Assist

Agents see the band and message. Nothing is enforced.

Ends when analysts agree with the band on sampled calls at the rate the bank sets.

Phase 3: Step up

Digit challenge and app prompt switch on for the top band. Analyst queue goes live.

Ends when confirmed takeovers fall and complaint rates stay inside the bank's limit.

Phase 4: Scale

More call types and languages. Retrain when drift alarms fire, not only on a calendar.

Reviewed each quarter by fraud-risk operations.

Where it plugs in

Audio from a SIP fork or the contact-centre platform's media stream (for example Amazon Connect or Genesys Cloud). The band goes to the agent desktop; cases go to the bank's case system.

Who owns what

Fraud-risk operations owns thresholds and cases. The CISO owns access and audit. IT owns the audio tap.

Friction budget

Clear adds no step. Step up adds six digits; sensitive requests always get it. Only Hold sends an app prompt.

Phases are a proposal for a pilot, not a deployed rollout.

Forty-eight hours at IIT Delhi

Hours 0 to 6	Hours 6 to 18	Hours 18 to 30	Hours 30 to 40	Hours 40 to 48
Call simulator, L1 and L2 running	Tune L2 on degraded audio; fusion, bands, digit challenge	Agent screen, context rules, app prompt	Analyst queue and evidence record	Held-out test, fixes, demo freeze

Live demo: three calls

Genuine caller	Clear. No extra step.	✓
Pre-recorded clone	Fails the digit challenge. Hold, app prompt, analyst case. The baseline attacker.	✗
Live voice changer	Main defence is L2. If it slips through, we show the miss and post-call review.	

Gathered beforehand

Public datasets, pretrained model weights and the consented Hinglish recordings.

Built on site

Integration, policy, screens, evidence record. If behind at hour 30, drop L4 and keep L1 to L3 and Hold.

Who does what

Pranjal Kumar: lead, security and policy

Sehaj Luthra: scoring models, L1 and L2

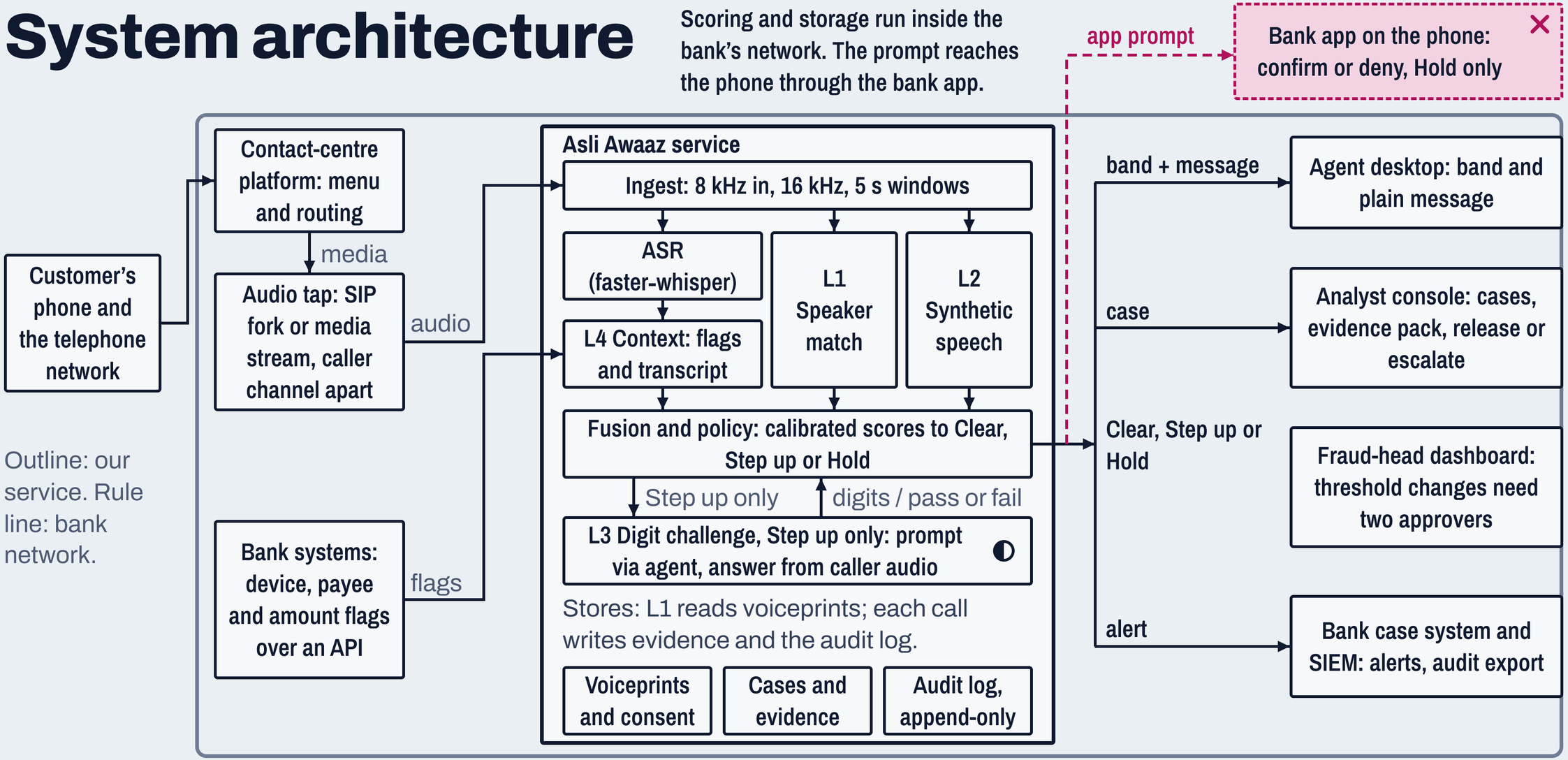
Yash Chauhan: screens and demo

Kushagra Raj: evaluation, red-team clones, GPU runs

L2 tuning and tests run on a laptop RTX 4060 GPU.

System architecture

Scoring and storage run inside the bank's network. The prompt reaches the phone through the bank app.



Outline: our service. Rule line: bank network.

Layer sizes and models: pitch page 5

Data, consent and attack surface

Input	Needed for	Consent or authorisation	Kept
Live call audio	Layers 1 to 3	Recording notice; DPDP basis, set by the bank	Held in memory during the call; written to disk only for Hold cases.
Voice embedding	L1 comparison	Opt-in in the bank app; withdraw any time	Until consent is withdrawn
Caller transcript	L4 urgency words	Same notice as audio	Held in memory during the call; written to disk only for Hold cases.
Bank flags	L4 context rules	Bank's own data, already held	Flags only, in the case record
Evidence clip	Analyst review of Hold cases	Role-gated; Hold cases only	Default 90 days, then deleted unless a case is open

Provenance. Datasets and models are listed with source and licence in Appendix A; model versions are pinned. Every score is logged with its model version.

What an attacker tries

Replay of a pre-made clone
Random digits on Step up; reply-time check.

Live voice changer
L2 on every window; post-call review if it misses.

Noise or codec tricks
Train on degraded audio; add new generators; red-team before a pilot.

Probing the system
Attempt caps; the caller learns only that an extra proof was asked, never which check fired or the score.

Misuse of voiceprints or evidence
Role-based access; encrypted with bank-held keys, TLS in transit; audit log.

Evidence-clip retention is the bank's policy

Escalation, review and performance

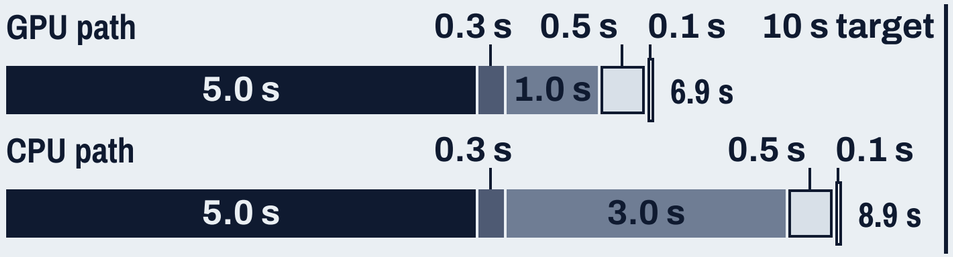


Role	Sees	Can do
Agent	Band and message	Carry on, read the digits, escalate
Customer	App prompt	Confirm, deny, report
Analyst	Scores, layer notes, Hold clip	Release, escalate, add notes
Fraud head	Dashboards, audit log	Change thresholds, two people

Evidence pack. Timestamps, band, per-layer scores, model versions, the Hold clip, actions and the analyst's decision, with a SHA-256 hash chained into the audit log.

Time budget per call

Target, not a result



- Audio window, 5.0 s
- Features, 0.3 s
- ASR for L4, 0.5 s
- L1 + L2, 1.0 s on GPU; 3.0 s on CPU with a weaker L2
- Fusion, 0.1 s

The budget sums to **about 6.9 s on GPU and 8.9 s on CPU**, before network and UI delay. The Step up outcome comes later. If scoring is late or fails: no band, normal process; on a sensitive request the agent runs the Step up digit challenge instead.

Validation, in order

1. Offline tests on degraded audio
2. Red-team with our own consented clones
3. Shadow mode on recorded calls from a bank
4. Limited Step-up pilot

Appendix: AI, data and third-party disclosures

Category	Item	Use and terms
AI-generated content	Research and drafting support from AI tools. The team checked each figure against the sources in Appendices C and D.	Pitch PDF
Datasets	ASVspoof 2019 LA and ASVspoof 5 (ODC-By); In-the-Wild (Apache 2.0 per Fraunhofer, CC BY-SA 4.0 per its Hugging Face card); planned Hinglish recordings, made with written consent that covers clone tests.	Training and testing
Models	SpeechBrain ECAPA-TDNN (Apache 2.0); wav2vec2 XLS-R 300M (Apache 2.0) with an AASIST head (MIT); AASIST-L (MIT); faster-whisper (MIT).	Pretrained weights; fine-tuning planned
Clone generators	Planned: OpenVoice V2 (MIT) and RVC voice conversion (MIT).	Test clones of consenting speakers, research use only
Tools and fonts	Python, PyTorch, ffmpeg. Fonts: Archivo, Archivo Narrow, Kalam, Rozha One, under the SIL Open Font License.	Build and layout
Data not used	No confidential bank, customer or platform data.	Figures are public or marked as targets

Appendix: assumptions we depend on

	We assume	How we test it, or who confirms
A1	**The bank can fork call audio from its platform, both sides apart.**	Integration talk with the bank; the prototype uses a recorded-call simulator.
A2	**8 kHz telephone audio keeps enough signal for L1 and L2.**	Error rate on degraded test audio, per codec and loss level.
A3	**Customers agree to voice enrolment; others use the app only.**	Consent flow design; opt-out rate is reported.
A4	**Public data and our Hinglish set stand in for fraud calls.**	In-the-Wild plus own set; gap to real calls reported in a pilot.
A5	**The bank can show a push prompt in its app within seconds.**	Dependency on the bank’s app team.
A6	**Compliance accepts the DPDP consent and retention design.**	Dependency on counsel review before any pilot.

Assumptions A1 to A6 and how each is checked

Appendix: sources (1 of 2)

#	Source	Used on
1	Lok Sabha Unstarred Question No. 344, written reply by MoS Home Bandi Sanjay Kumar, 22 Jul 2025: ₹22,845.73 crore (2024), ₹7,465.18 crore (2023), 22,68,346 cybercrime incidents on NCRP (2024). mha.gov.in/MHA1/Par2017/pdfs/par2025-pdfs/LS22072025/344.pdf	Page 2
2	J. Cox, “How I Broke Into a Bank Account With an AI-Generated Voice”, Vice Motherboard, Feb 2023 vice.com/en/article/dy7axa	Page 2
3	Müller et al., “Does Audio Deepfake Detection Generalize?”, Interspeech 2022; In-the-Wild dataset card, CC BY-SA 4.0 arxiv.org/abs/2203.16263	Pages 6, 7
4	RBI, Master Directions on Fraud Risk Management in Commercial Banks (incl. RRBs) and All India Financial Institutions, 15 Jul 2024 rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=12702	Page 9
5	Digital Personal Data Protection Act, 2023	Pages 9, Appendix B

Appendix: sources (2 of 2)

#	Source	Used on
6	ASVspoof 2019 and ASVspoof 5; model cards for the models in Appendix A datashare.ed.ac.uk/handle/10283/3336 zenodo.org/records/14498691	Page 7, Appendix A
7	ICICI Bank press release, 25 May 2015: voice recognition for call-centre banking icici.bank.in/about-us/news-room/2015/news-icici-bank-introduces-voice-recognition-for-biometric-authentication	Page 2
8	Pindrop, “Pindrop Pulse for audio deepfake detection”; vendor claims, not tested by us pindrop.com/article/pindrop-pulse-for-audio-deepfake-detection/	Page 6
9	Amazon Connect live media streaming; Genesys Cloud AudioHook Monitor docs.aws.amazon.com/connect/latest/adminguide/customer-voice-streams.html help.genesys.cloud/articles/audiohook-monitor-overview/	Page 9

Appendix: where each criterion is answered

Criterion	Points	What the brief asks for	Where to look
Problem fit and user understanding	20	Primary user, threat scenario, harmful moment, scope	**Page 2; journey on page 3**
Technical architecture and detection logic	20	Layers, what each misses, bands, system diagram	**Page 5; architecture pages 1 to 3**
Solution value and differentiation	15	Before and after, what the agent sees, versus benchmark detectors	**Pages 3, 4 and 6**
Feasibility and finale execution plan	15	Data, validation plan, assumptions, 48-hour plan and roles	**Pages 7 and 10; Appendix B**
Safety, privacy and responsible AI	15	Five trust questions, fairness, consent, attacks, evidence path	**Page 8; architecture pages 2 and 3**
Adoption, integration and scale	10	Integration point, owners, rollout phases, friction, latency	**Page 9; architecture page 3**
Communication quality	5	Sources and disclosures	**Appendices A, C and D**

Final check from the brief. 01 Problem: page 2. 02 Product, input to evidence record: pages 3 and 4. 03 Architecture: architecture page 1. 04 Proof plan: pages 7 and 10, Appendix B. 05 Safeguards: page 8, architecture pages 2 and 3.